

ADVANCED

Statistical Analysis Cheat Sheet

The statistics concepts that separate 'I made a chart' from 'I can defend this conclusion.' Written for analysts, not statisticians.

1. Describing a Dataset

Concept	What it tells you	Rule of thumb
Mean vs. Median	Mean is sensitive to outliers, median isn't	If mean and median differ a lot, check for outliers or skew
Standard Deviation	How spread out values are from the mean	~68% of data falls within 1 SD of the mean (normal distribution)
Interquartile Range (IQR)	Spread of the middle 50% of data — outlier-resistant	Outlier if value is beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$
Skewness	Whether data leans left or right of center	Right-skewed (long tail right) is common for income, prices, wait times

2. Correlation

Concept	What it tells you	Rule of thumb
Correlation coefficient (r)	Strength & direction of a linear relationship, from -1 to +1	$ r > 0.7$ is strong, 0.3–0.7 moderate, < 0.3 weak
Correlation $\neq$ Causation	Two variables moving together doesn't mean one causes the other	Always ask: could a third factor explain both?

Classic trap: ice cream sales and drowning deaths correlate strongly — both are driven by hot weather, not each other.

3. Hypothesis Testing (in plain English)

Concept	What it tells you	Rule of thumb
Null hypothesis (H0)	The 'nothing changed / no effect' assumption you're testing against	e.g. 'The new checkout page has no effect on conversion rate'
p-value	Probability of seeing your result (or more extreme) if H0 were actually true	$p < 0.05$ is the common (not universal) threshold to reject H0
Statistical significance	The result is unlikely to be due to random chance alone	Significant $\neq$ important — check the effect size too

Concept	What it tells you	Rule of thumb
Confidence Interval	A range likely to contain the true value, at a stated confidence level	A 95% CI means: if you repeated this study, 95% of intervals would contain the truth

4. Choosing a Test

Question you're asking	Test to use
Do two group means differ? (e.g. Version A vs B revenue)	t-test (independent samples)
Do more than two group means differ? (e.g. 4 regions)	ANOVA
Is there a relationship between two categorical variables?	Chi-square test
Does a numeric variable predict another numeric variable?	Linear regression
Before/after comparison on the same subjects	Paired t-test

5. Regression, Quickly

Concept	What it tells you	Rule of thumb
R-squared (R²)	% of variance in the outcome explained by the model	Higher isn't always better — watch for overfitting with too many variables
Coefficient	How much the outcome changes per 1-unit change in a predictor	Check the sign (+/-) and whether it makes business sense
p-value per coefficient	Whether that specific predictor is statistically significant	A predictor with $p > 0.05$ is often not worth keeping in the model
Multicollinearity	When predictor variables are highly correlated with each other	Makes coefficients unstable and hard to interpret — check with VIF

6. Five Traps Analysts Fall Into

- **Sample size too small** — a 5% lift on 40 users isn't a finding, it's noise.
- **P-hacking** — testing 20 metrics until one hits $p < 0.05$ by chance is not a discovery.
- **Survivorship bias** — analyzing only customers who stayed ignores why the others left.
- **Simpson's Paradox** — a trend can reverse when you split data into subgroups. Always check segments.
- **Confusing 'significant' with 'big'** — a huge dataset can make a tiny, meaningless effect 'significant.'

Terence Rebello · The Analyst's Toolbox · terencerebello.com